

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a039d2969ec31b9bc.jsonl`  
- SHA-256: `7864616570f7c4750ec79b619c648992fa4740695aab5148f7142326b3d9b14d`  
- Source modified: `2026-07-02T03:05:33+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T03:02:48.192Z)

You are an adversarial verifier in a tech-debt audit. Your default stance: REFUTE. Today is 2026-07-02.
A finder claims the following about the repo at C:/Users/avidu/Projects/khelsutra-guru/khelsutra (avidullu/khelsutra):

```
{  
  "title": "Issue #114 stale-open: CF Access guest-login debugging against a deployment that no longer exists",  
  "category": "hygiene",  
  "severity": "P2",  
  "file": "deploy/DEPLOY_ALPHA.md",  
  "line": 105,  
  "evidence": "The issue is pure Cloudflare-dashboard triage (allowlist policy on the Access app per deploy/DEPLOY_ALPHA.md:105) for the droplet alpha, deferred by owner 2026-06-23. The droplet was retired 2026-06-25 and alpha is paused (memory session-handoff.md:19), so there is nothing behind app./api. to sign into; the Access app + dangling CNAMEs are themselves pending cleanup. Nothing in-repo can fix or verify it.",  
  "proposed_fix": "Owner call: either close as overtaken-by-events with a note, or retitle to 'verify a second allowlisted email can sign in' as a go-live checklist item for the Cloud Run alpha (the 5-point debug checklist in the issue body is still the right procedure then). No code change.",  
  "effort": "S",  
  "tracked_in": "khelsutra#114"  
}
```

Repo root candidates: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, C:/Users/avidu/Projects/khelsutra-guru/khelsutra, C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector, C:/Users/avidu/Projects/khelsutra-guru/rally-annotator, C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport (pick from the finding's repo field; non-git dirs: 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup', C:/Users/avidu/Projects/khelsutra-guru/khelsutra-screenshot).

VERIFY against the CURRENT working tree (just pulled to origin master/main): (1) does the cited code actually exist and behave as claimed at file:line? (2) was it already fixed by a recent commit (git log --oneline -15 -- <file>)? (3) is the severity honest - construct the concrete failure scenario or downgrade; (4) is the proposed fix sane and correctly scoped; (5) is it owner-gated per: alpha/serving go-live, Studio P10/P11 corpus-promotion/training, product/strategy/licensing, real cloud spend, counsel ToS/DPA?

HARD RULES: READ-ONLY (no file modifications, no installs, no test-suite runs). gh read-only commands allowed. Your reasoning MUST cite file:line you actually read. CONFIRMED requires you personally saw the defect/debt in current code; if you cannot reproduce the claim from the code, say REFUTED; if a commit fixed it, ALREADY_FIXED.

2. user (2026-07-02T03:02:59.552Z)

Already up to date.

f9eea14 web: resilient video-summary readers + show fps in the library (#140)
9223ba9 docs: refresh khelsutra docs for today's storage-track work (P2-P5) (#139)
fb2f561 web: persistent "FFmpeg not detected" banner (hard-blocker warning) (#138)
8667dbf P5: SPA storage-medium picker (store a video into your medium) (#137)
a04a21b docs(tracker): inline the D14 decisions into the P2-P5 rows (#136)

3. user (2026-07-02T03:03:06.098Z)

```
{"body":"## Symptom (real demo, 2026-06-23)\n\nDemoing the alpha (`app.khelsutra.guru`) to a friend: his email was whitelisted in the Cloudflare Access policy, but he could not sign in (the email-OTP / Access login didn't let him through), so he couldn't use the app with his own address. The owner's own email works fine.\n\nThe core flow is great ? small clip ? highlights was \"pure bliss.\" This is purely the Access allowlist not admitting a second, whitelisted address.\n\n## Context (what's deployed)\n\n- CF Access app covering `app.` + `api.khelsutra.guru`, team `khelsutra.cloudflareaccess.com`, email-OTP login gated to an allowlist policy.\n- Per the deploy notes the policy was created allowing a single email (`avi.dullu@gmail.com`); adding a guest means the guest's email must be in that app's policy include rule.\n- Background: `deploy/DEPLOY_ALPHA.md` (the Access/tunnel setup).\n\n## Likely causes to check (in order)\n\n1. Was the email added to the right policy? The include rule must be on the Access application that covers `app.khelsutra.guru` (not a different app), and the policy action must be Allow.  
2. Include selector type ? \"Emails\" (exact addresses) vs \"Emails ending in\" (domain). A typo / trailing space / case in the exact-email entry will silently fail to match.  
3. Policy actually saved + propagated ? confirm the edit was saved on the app and re-test in a fresh/incognito session (Access caches the session cookie).  
4. OTP delivery ? did the friend receive the one-time PIN email at all? (spam folder / wrong address / provider blocking).  
5. Single-origin note ? the working path is `api.khelsutra.guru` (the engine serves the SPA single-origin); confirm the guest used that, not a stale `app.` Pages URL with the cross-origin cookie issue (see issue history).\n\n## Repro\n\n1. Add `guest@example.com` to the Access app's Allow policy (Emails ? exact).  
2. From a clean browser, open the app URL ? enter `guest@example.com` ? request OTP.  
3. Expected: PIN email arrives ? entering it admits the guest. Actual: login fails / not admitted.\n\n## Severity / priority\n\n? Not blocking (owner can use it; demos work by driving it yourself) but blocks sharing with whitelisted guests ? needed before inviting alpha testers. Deferred by owner 2026-06-23.\n\n,\"comments\": [], \"createdAt\": \"2026-06-23T10:25:52Z\", \"labels\": [], \"state\": \"OPEN\", \"title\": \"Cloudflare Access: an allowlisted email still couldn't sign in (alpha auth)\", \"updatedAt\": \"2026-06-23T10:25:52Z\"}
```

4. user (2026-07-02T03:03:06.223Z)

```
80 python -m backend.main --server # binds 127.0.0.1:8000 ? kept loopback by RALLY_BIND_HOST ($2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
81 ```
82 Keep it alive with a process manager (`systemd` on the droplet, mirroring `site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
83
84 ---
85
86 ## 3. The SPA on Cloudflare Pages (the `app.` side)
87
88 The SPA ships as a static Pages site, git-connected like the marketing site (auto-deploy on merge to `master`).
89
90 1. Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git ? pick the `khelsutra` repo.
91 2. Build settings:
92 - Root directory: `web`
93 - Build command: `npm run build`
94 - Build output directory: `web/dist`
95 - Environment variable: `VITE_API_BASE = https://api.khelsutra.guru/api` ? this is what points the SPA at the engine tunnel (PR #64).
96 3. Custom domain: add `app.khelsutra.guru` to the Pages project.
97
98 > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first, Pages builds a per-PR preview URL.
```

```

99
100    ---
101
102    ## 4. Cloudflare Access (gate BOTH hostnames)
103
104    1. **Zero Trust ? Access ? Applications ? Add ? Self-hosted.**
105    2. **Application domain:** add **both** `app.khelsutra.guru` **and**
    `api.khelsutra.guru` to the *same* application (one app ? one shared session cookie, so a tester
    signs in once and both the SPA and its API calls are authorized).
106    3. **Policy:** Action *Allow*, rule *Emails* ? your alpha testers' addresses (free ? 50
    users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
107    4. **Copy the application `AUD` tag** ? paste into the engine's `config.json`
    `security.cf_access_aud` ($2.2), and set `cf_team_domain` to `https://<your-
    team>.cloudflareaccess.com`. Restart the engine.
108    5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
    `api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
    options requests to CORS preflight?** (or add an explicit OPTIONS bypass), else the preflight is
    redirected to the login page and every API call fails. The engine's CORS middleware
    (`allow_credentials=false`, no cookies) then answers the preflight.
109
110    ---
111
112    ## 5. The tunnel (engine ? Cloudflare)
113
114    ```bash
115    cloudflared tunnel login                                # browser auth to your zone
116    cloudflared tunnel create khelsutra-studio              # note the <TUNNEL_UUID>
117    # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
    <TUNNEL_UUID>
118    cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru
119    cloudflared tunnel run khelsutra-studio                  # or install as a service:
    `cloudflared service install`
120    ```
121    See `deploy/cloudflared.config.example.yml` for the ingress (only `api.khelsutra.guru` ?
    http://127.0.0.1:8000).
122
123    ---
124
125    ## 6. Verify (the posture test + the CUJ)
126
127    **Exposure posture (must hold ? this is what makes it safe to share):**
128    - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
    **refused / timeout** (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
    directly reachable.
129    - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
    the Access login** (not a 200). ? The gate is in front.
130    - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
    SPA loads and `/api/health` returns 200 (same session cookie). ?
131
132    **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
    **upload a video from their device** (the SPA chunked-upload lands it in `upload_dir`, inside
    the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All
    from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
    `allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
133
134    ---
135
136    ## 7. Honest caveats / deferred
137
138    - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
    upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
    Keep `upload_dir` inside the jail ($2.2).
139    - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
    `allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
    that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`

```

(pinned by engine `tests/test\_api\_endpoints.py::test\_process\_hosted\_mode\_rejects\_nonlocal\_uri`). Cloud-URI ingest works **only** when `allowed\_video\_root` is unset (the local single-user posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine change, not a config flip.)*

5. user (2026-07-02T03:03:12.859Z)

```
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? badminton-highlight-indexer (rename PENDING) + siblings: sports-
data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-25 (**review-automation + audit-completion session** ? see ?
You are here (2026-06-25) below. Maintenance/PR-review phase, NOT serving/alpha: 0 open PRs, all
CI green, no GCP VMs, PR-review watch loop OFF. The serving/alpha track below is UNCHANGED +
owner-gated.) Prior: 2026-06-22 (**M11+M12 serving-code session** ? owner: "build 3 then 4 then
we pair on the alpha (2)". **Both Claude-doable engine items DONE + merged:** **M11** data-
flywheel skeleton [#336](https://github.com/Khelsutra/badminton-highlight-indexer/pull/336)
(`334ad0f` ? default-OFF + consent FAKED-DENY, inert; reuses workspace/eval/promotion; NO
migration) + **M12** `gdrive-api://` OAuth StorageBackend
[#338](https://github.com/Khelsutra/badminton-highlight-indexer/pull/338) (`819e395` ?
idempotent put, lazy Google imports, injected-fake tests; live per-user OAuth owner-gated). Both
adversarially reviewed, full-suite+cross-OS green, merged on green, in an isolated worktree. **?
NEXT = pair with owner on the ALPHA GO-LIVE (option 2)** ? owner brings CF dashboard + box +
rotated Gemini key; the counsel ToS/DPA inputs (the 7-pt checklist) gate M11 *live* +
M9-consent. ? schema ladder drifted under me this session: v7=`locale`, v8=compute-picker ? the
future M9-consent migration is **v9** (check `PRAGMA user_version` before authoring). Prior
same-day: **maintenance / green-PR merge-sweep session** ? owner asked to merge the green PRs +
finish loose ends. 6 PRs now on master: worker dated-label fix **#319**, nightly Cloud-Run
scheduler **#320**, DATA_IN_GCS map **#316**, doc **#321**, gate-A litmus re-ground **#322**,
golden-fixtures backlog **#323**. engine `origin/master` = `9b37269`, suite **14287**, **0 open
engine PRs**; docs-archive had no terminal candidate (owner-gated). Full per-PR detail =
[[current-state]] top. ? ~10 concurrent worktrees/sessions ? owner is trimming concurrency
before new work. The serving/alpha plan below is UNCHANGED. Prior **2026-06-21 ALPHA-SHAREABLE
session** ? Claude's half of "make it UI-driven + alpha-shareable" is **DONE**: 3 PRs merged ?
engine **#294** exposure-safety boot posture + khelsutra **#64** SPA `VITE_API_BASE` + **#65**
deploy kit. **KEY CORRECTION:** B-P2 (SPA ingest screen) was ALREADY shipped (#56) ? alpha-
shareable was a DEPLOY/AUTH task, not an SPA build. Owner chose the **SPLIT** topology (SPA on
CF Pages + `api.` cloudflared tunnel + CF Access). Remaining = OWNER stand-up via
`khelsutra/deploy/DEPLOY_ALPHA.md`; then engine M11/M12. Full record [[ui-driven-cloud-serving-
gap]]). Prior: HUGE serving session ? M1?M9-auth shipped (#279?#291) + real M7 L4 run. **Repo is
under the `Khelsutra` GitHub org**.
13
14     > **Read order:** this file ? [[current-state]] top (dense per-checkpoint detail) ?
`docs/NEXT_STEPS.md` (the prioritized cross-track plan, refreshed 2026-06-21) ?
`docs/DOC_STATUS.md` (doc lifecycle/hygiene).
15
16     ## ? You are here (2026-06-25)
17     **Review-automation + audit-completion session.** engine `origin/master` = **0b4f492`**
. all CI green . **0 open PRs** . no GCP VMs . **PR-review watch loop OFF** (cancelled). NOT
serving/alpha (that track is the "Earlier" block below, unchanged + owner-gated).
18     - **Repo tidy:** synced master, pruned ~92 stale local branches + 7 stale worktrees;
**fixed a corrupted (UTF-16-tail) `.gitignore`** ? `artifacts/` ignored +
`KHELSUTRA_PRODUCT_OVERVIEW.md` tracked (**#377**); owner un-hid the hidden
`C:\Users\avidu\Projects` folder.
```

19 - ****khelsutra site-droplet RETIRED**** ([[khelsutra-site-droplet-parity]]): the dual-purpose `claude-do-rally-test`/`168.144.126.176` (marketing-site mirror + alpha engine) kept drifting from CF ? ****deleted (\$0; data archived to `gs://khelsutra-rally-corporis/archive/droplet-?-20260625/` + local)**, parity monitor removed (khelsutra #117), Node-20 actions bumped (#118). **** ? This means the prior "LIVE alpha on the droplet" in the Earlier block is ****GONE**** ? alpha is paused pending the Cloud-Run move; `api.khelsutra.guru` CNAME/CF-Access need cleanup.**

20 - ****CODE_CLEANUP_AUDIT advanced ? 3 HIGH items + infra all merged:**** B2/P17 surface-Gemini-failures (****#378****) . 01/P11 Gemini remote-file GC (****#380****) . B1/P9 remove-fabricated-motion-data (****#385****); + ruff-format split (#379), audit-doc reconcile/accuracy (#375/#376), Tier-1 nightly `nightly.yml` (#381), per-job CI timeouts (#382). ****Remaining audit items: D1 (split `main.py` ~1450 lines), D13 (ByteTrack?`trackers`), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/O3/O4).**

21 - ****NEW tracked project: Decision-Snapshot Regression**** ([[decision-snapshot-regression]] . `docs/DECISION\_SNAPSHOT\_REGRESSION.md` . ****#383 merged****) ? a deterministic ****quality-neutral proof**** (freeze+compare the DECISION output = segments + stitch plan off a cached trajectory; ****NOT**** a reel-MD5). ****Decisions D1?D11 LOCKED**** (D10 = double-run determinism guard, deterministic segmenters only; D11 = advisory?gating staging). **** ? NEXT = \$7 P1**** (canonical snapshot schema/serializer) ? P2 (`build\_stitch\_plan()` extract) ? P3 (extend `test\_golden\_fixtures` + guard). B1/#385 makes `motion` snapshot-eligible.

22 - ****Review-automation patterns established**** ([[review-signoff-policy]] + [[review-trust-ci-over-rerun]]): spin off ****multi-agent review Workflows**** (Ultracode; agents READ-ONLY) ? adversarially verify ? post synthesized findings; ****sign-off policy**** = LGTM pure-refactors on green (AST-identity vs master), HOLD logic/CI-infra for owner; ****written LGTM only**** (gh=`avidullu` = owner's own account ? GitHub blocks self-Approve); ****trust green CI + a PR `## Verification` section, don't re-run identical CI checks**** ? reserve local effort for what CI can't cover (AST-identity, merge-safety, env-noise-vs-real-defect, findings-actually-fixed).

23

24 **** ? Likely next pick-up:**** start the Decision-Snapshot project at ****\$7 P1****, or continue the audit (D1 split-main.py / D13 ByteTrack). Serving/alpha remains owner-gated (now needs the Cloud-Run move since the droplet is gone).

25

26 ---

27

28 **### Earlier ? serving/alpha track (2026-06-22/23, UNCHANGED + owner-gated; note the droplet alpha is now retired per above)**

29

30 **## ? You are here (2026-06-22)**

31 ****(2026-06-23 #342)**** engine `origin/master` = ****`688d28f`**** ? merged a small compute-path fix: `GoogleCloudRunJobClient.run\_job` now guards an unset/empty `GOOGLE\_CLOUD\_PROJECT` (was passing `Optional[str]` into `JobsClient.job\_path(project: str,?)` ? latent mypy `arg-type` seen only with `google-cloud-run` installed + a bogus `projects/None/?` path at runtime). Guard runs before the lazy SDK import (CI-testable, SDK-absent) and raises a clear `RuntimeError`; +`test\_real\_client\_rejects\_missing\_project`. Full suite 1488?, cov 85.86%, all 10 CI green, merged on green per [[working-agreement-merging]]. Follow-up doc [#345](https://github.com/Khelsutra/badminton-highlight-indexer/pull/345) (`9329473`) added the operator note to `deploy/cloudrun/README.md` Gotchas. (Originated as a spawned chip; the live cloud-toggle work above is the sibling track ? see [[current-state]] top.)

32

33 ****(2026-06-22 M11+M12)**** engine `origin/master` = ****`819e395`**** (past #336 M11 + #338 M12; also sibling #333 M6?/api/process, #334 Cloud-Run env, #6d3d054 SPA compute-picker) . khelsutra `origin/master` = `ba38f15` (past #96 Privacy Policy, #100). ****The two Claude-doable serving-code items are DONE**** (M11 flywheel skeleton #336 + M12 gdrive-api #338 ? both default-OFF/inert, adversarially reviewed, merged on green). ****Claude's serving + alpha-shareable code is now complete.**** **** ? THE ONLY REMAINING ALPHA STEP IS OWNER-DRIVEN: pair on the go-live**** (cloudflared tunnel + CF Pages + CF Access + Gemini-key rotation per `khelsutra/deploy/DEPLOY\_ALPHA.md`) ? Claude guides, owner drives the CF dashboard/box. M11 going *live* + M9-consent need the counsel ToS/DPA inputs (the 7-pt checklist in [[ui-driven-cloud-serving-gap]] / this session). Other future engine code: M7 real cloud-serving e2e (owner GCP spend), per-video workspace P1?P6, the rally-quality track (corpus-gated).

34

35 ****(2026-06-22 sweep)**** engine `origin/master` = ****`9b37269`**** (+ today's #316/#319/#320/#321/#322/#323) . engine ****~1428**** tests green (combined-master verified w/ corpus manifest) . ****0 open engine PRs**** . no GCP VMs. This was a cross-track maintenance sweep (worker fix, nightly scheduler, GCS-data doc, gate-A litmus re-ground, golden-fixtures backlog)

? the serving/alpha forward-plan below is unchanged. ? a sibling's redundant gate-A branch
`fix/served-gate-a-regrounded-15clip` sits in the MAIN checkout; concurrency being trimmed.

36

37 ******(2026-06-21 prior)****** engine `origin/master` = `0c0e064` + (past #292/#293/#294) .
khelsutra `origin/master` = `4627830` (past #63/#64/#65) . ****no GCP VMs (audited \$0)**** . engine
~1259 tests green.

38 *******? ALPHA-SHAREABLE SESSION (2026-06-21) ? 3 PRs MERGED; Claude's half of the owner's
top priority is DONE:******

39 - ****KEY CORRECTION** (anti-spiral [[lesson-verify-engine-surface-before-scoping]]):****** the
handoff said "do the B-P2 SPA ingest screen + M7 Cloud Run *service*" ? but ****B-P2** was already
shipped****** (khelsutra #56, live e2e in `DRY\_RUN\_CUJ.md` §9) and recon showed alpha-shareable is a
****DEPLOY/AUTH**** task, not an SPA build. CORS is already env-driven; the engine needed no CORS
code.

40 - ****Owner decisions** (this session):****** (1) pursue alpha-shareable via topology ******(i)
Tunnel-to-a-box******; (2) sub-topology = ****SPLIT**** (SPA on CF ****Pages**** + engine on `api.`
****cloudflared tunnel**** + CF Access over both ? engine `HOSTING\_PLAN.md` §2), single-origin Caddy
kept as the alternative.

41 - ****engine** [#294](https://github.com/Khelsutra/badminton-highlight-indexer/pull/294)
(`0c0e064`) ? exposure-safety boot posture:****** `backend/config/startup.py` fails fast at boot
(server-only `include\_exposure=True`) when `security.edge\_auth\_enabled` is ON but half-
configured ? `EXPOSED\_NO\_VIDEO\_ROOT` (fatal, jail unset), `EXPOSED\_EDGE\_AUTH\_INCOMPLETE` (fatal,
CF team/AUD), `EXPOSED\_AUTH\_EXTRA\_MISSING` (warn). ****Default-OFF**** (edge_auth off ? no-op;
worker never opts in). Aligned `auth.resolve\_tenant` whitespace-strip. Adversarial 3-lens review
(`wf\_35b735a9`) ? no bypass. +16 tests; 1259 green cross-OS.

42 - ****khelsutra** [#64](https://github.com/avidullu/khelsutra/pull/64) (`8bbc557`) ? SPA
`VITE\_API\_BASE`:****** `apiBase()` env-driven (default same-origin `/api`) + `vite-env.d.ts` + 3
tests; build-smoke proved the override bakes in.

43 - ****khelsutra** [#65](https://github.com/avidullu/khelsutra/pull/65) (`4627830`) ? alpha
deploy kit:****** `deploy/DEPLOY\_ALPHA.md` (full split runbook + CORS-preflight gotcha + posture
test + deferred notes) + `deploy/cloudflared.config.example.yml` + README/tracker pointers.

44 - ****Discipline****: isolated PRs off `origin/master` (khelsutra in a dedicated
****worktree**** ? main tree had sibling A8 WIP, since merged as #63), default-OFF, adversarial
review, full CI green cross-OS, merged own green PRs, tracker+changelog in-PR. ****?** REMAINING for
alpha = OWNER stand-up only****** (tunnel + Pages + CF Access + key rotation per
`khelsutra/deploy/DEPLOY\_ALPHA.md`). ****M1?M9-auth** (prior sessions) detail below.******

45 *******? HUGE SERVING SESSION (2026-06-21) ? 6 PRs MERGED + a REAL cloud-GPU run, owner
BATCH-AUTHORIZED the default-OFF code path + \$100/?10k spend:****** ****#285**** (docs §8 D16/D17 + P3
reconcile) . ****#286 M5**** (`backend/config/startup.py` per-profile fail/warn-fast checks + the
****L4 profile-parity**** proof; worker stays torch-free) . ****#287 M6**** (`backend/compute/`
ComputeBackend registry ? `CloudRunCompute` dispatch a Cloud Run ****Job**** + bucket-poll;
`deploy/cloudrun/{Dockerfile,README}`; mocked) . ****#288 M8**** (per-job ****cost ledger**** + v6
migration + versioned `pricebook` \$0 + `/cost` API) . ****#290 M9-auth**** (Cf-Access JWT verify
RS256 + `owner\_id` tenant tag; security review found NO bypass; alg=none + HS256-confusion
locked) . ****#291 ops-agent codify** + `docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`******
(`deploy/gcp/create\_gpu\_vm.sh` always bakes the Ops Agent). All ****default-OFF****, isolated,
****adversarial-reviewed** (a workflow per PR)******, full-suite+cross-OS CI green, tracker+changelog
in-PR. ****?** REAL M7 L4 RUN DONE + VERIFIED:****** native WASB on a real L4 ? ****22 rallies**** on the
smoke clip ? GCS out; VM deleted (\$0, ~\$0.21). ? the no-Gemini cloud run needs
****`skip\_ai\_handoff=true`**** (the long-flagged "0 rallies" cold-start fix). A 2nd dry run via
`create\_gpu\_vm.sh` ****confirmed Ops-Agent auto-installs**** + found the ****watermark-font gap****
(fixed in #292: `bootstrap.sh` now installs `fonts-dejavu-core`). ****M1?M4** (prior session) detail
below.******

46 - ****M1** ([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279),
`ce22791`):****** `deployment.profile`+`compute\_target` on `AppConfig` + `backend/config/presets.py`
(PROFILE_PRESETS + `suggest\_profile`) + loader preset/precedence
(defaults<preset<config.json<env via `deep\_merge`; no-profile path = exact old shallow merge) +
auto-detect-as-SUGGESTION (D15). Pure/additive, profile='' == today. 3 minor review findings
folded.

47 - ****M2** ([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280),
`e111add`):****** `StorageConfig.input\_backend`/`input\_root` +
`backend/storage/{ref.resolve_input_uri,
__init__.resolve_input_ref/acquire_local_input/input_is_local}` + `main.py` API/CLI wired
(acquire local; keep source URI for DB/report). ****local = identity passthrough** (byte-for-byte
unchanged)******; bucket/Drive fetched to scratch (mirrors worker `cmd\_infer`). Unknown scheme?clean
400 (was a 500 regression ? caught by review); `local://` accepted on resolved key; hosted-mode

```

rejects non-local; OS-independent anchored-path detection (os.path.isabs differs Win+py3.13). 1
major + 6 minor review findings folded. Suite 1105 green, cov 84.84%.
48 - **M3 ([#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282),
`2afd54f`):** configurable output/scratch base ? `backend/utils/paths.output_base(config)`
(returns `output.output_dir`, default `"output"`); routed all 6 hardcoded
`os.path.join("output", ?)` scratch sites in `trajectory_hybrid`/`yolo_hybrid`/`gemini` through
it (create+cleanup same var). **DEFAULT `"output"` ? byte-identical** (golden + cross-OS green);
a custom `--output-dir` now actually moves scratch (latent bug fixed). Default-OFF ? no Launch
Report. `tests/test_output_base.py` + a yolo call-site capture; clean-bill focused review (yolo-
coverage minor folded). Suite 1114 green.
49 - **M4 ([#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283),
`cee4baa`):** `DeviceContext` (`backend/pipeline/detectors/device.py`) + native WASB **CPU-
fallback** via a new `device="auto"` (cuda-if-available-else-cpu). **Explicit `"cuda"` stays
STRICT** (healthcheck still hard-fails without a GPU ? no silent slow-CPU downgrade; "cuda
default unchanged"). CUDA probe **cached** (healthcheck?run_predict reuse);
`_effective_device()` resolves before argv so "auto" NEVER reaches `wasb_infer.py`; fail-fast
on a typo'd device. `wasb_infer.py` already supports cpu ? no change there. Device-selection
tested offline (mocked subprocess); **real CPU-inference run = owner/GPU-verified.**
`tests/test_device_context.py` + native-runner additions; clean-bill focused review (2 nits
folded). Suite 1126 green, cov 84.96%.
50 ***? NEXT (priority order):**
51 1. **? ALPHA-SHAREABLE ? Claude's half DONE this session (#294/#64/#65).** Engine is
exposure-safe by construction; SPA targets a split origin; the deploy kit + runbook are merged.
**? REMAINING = OWNER stand-up** (not Claude-doable): on the box `pip install -e .[auth]` + the
safe-exposed `config.json` + `RALLY_CORS_ALLOW_ORIGINS` + rotate `GEMINI_API_KEY` + run
`backend.main --server`; CF ? cloudflared tunnel (`api.khelsutra.guru`) + CF Pages
(`app.khelsutra.guru`, `VITE_API_BASE`) + ONE CF Access app over both + the OPTIONS-preflight
bypass. **Runbook = `khelsutra/deploy/DEPLOY_ALPHA.md`.** *(A later UI polish, not the alpha
blocker: a compute PICKER in the UI, D13/D15.)* Full record: [[ui-driven-cloud-serving-gap]].
52 2. **M11** (data-flywheel plumbing) ? ? privacy-sensitive (consent/data), give it FRESH
focus: `backend/flywheel.py` capture?store?shadow-eval?goldenize reusing
`workspace.py::for_video` + `backend/eval/experiment.compare` +
`backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked
`consent_attested`. ? depends on M9's not-yet-landed **v7 consent migration** (M8 took v6) +
counsel ToS/DPA; only a default-OFF skeleton is Claude-doable.
53 3. **M12** (`gdrive-api` OAuth StorageBackend) ? `backend/storage/gdrive_api.py`
`GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex allows the hyphen; add to
`_KNOWN`+`_backend_class`+the StorageConfig Literal) vs an INJECTED fake Drive (mirror `gcs.py`
DI); keep mount-only `gdrive://`. ? M9-consent cols + per-video P1 = a new **v7** migration (M8
took v6).
54 ***? owner-only residual:** real M5 truly-local runlog (template pre-staged) · GCP spend
for the M7 *service* deploy + a real e2e · counsel ToS/DPA (M9-consent) · real GCP prices+FX +
the M8 usage-emission follow-up · Google OAuth app (M12 real) · CF team-domain/AUD+ingress-lock
· repo rename · DO droplet/token retire.
55
56 ## ? Icebreaker for the NEXT session (copy-paste)
57 > Continue the badminton-highlight-indexer ENGINE serving + UX track. FIRST: `git pull
--ff-only`, then read memory `session-handoff.md` + `current-state.md` (top) + [[ui-driven-
cloud-serving-gap]] + [[compute-stack-gcp-only]].
58 > STATE: **The cloud alpha is LIVE + proven end-to-end** (2026-06-22 night: a real match
? **27 rallies ? reel downloaded**, on the droplet `claude-do-rally-test`/`168.144.126.176`
behind CF Access at `app.khelsutra.guru`; fixed live: `app.` DNS?the tunnel, the Gemini key,
`RALLY_ALLOWED_HOSTS`). **Engine #294 + M11 #336 + M12 #338, khelsutra #64/#65 all merged.**
59 > ***? TOP NEXT ACTION ? owner LOCKED IN the GCP L4 (2026-06-22 night): spin up the L4
for the NON-GEMINI phase** (native WASB inference + owned-model training, NO Gemini). Run
**`bash deploy/gcp/create_gpu_vm.sh`** (defaults `RALLY_GPU=l4` ? `g2-standard-4` 24GB, zone-
sweeps Mumbai?Singapore, bakes Ops-Agent) ? pull corpus from `gs://khelsutra-rally-corpus`
([[gcs-data-source-of-truth]]) ? `python -m backend.worker infer|train` with
`skip_ai_handoff=true` ? **`bash deploy/gcp/teardown.sh delete` AFTER** ([[gcp-vm-always-
delete]]), **$100/?10k cap**. On-demand ~$0.74/hr Mumbai (?$1?2/interactive session, fine);
`RALLY_GPU=t4` fallback if L4 stocks out; add a Spot toggle (`--provisioning-model=SPOT`,
currently STANDARD at `create_gpu_vm.sh:41`) before any LONG training run. Full config/pricing =
[[compute-stack-gcp-only]].
60 > **Also queued ? 3 alpha-hardening issues from the live run (Claude-doable, default-
safe):** khelsutra#111 (SPA survive-refresh + live progress ? the 10-min poll ceiling timed out

```

~secs before a long job finished), khelsutra#112 (WhatsApp share the VIDEO via Web Share API, not just text), engine#340 (idempotent `api/process` ? no Gemini RE-BILL on refresh/"Try again").

61 > Discipline: ONE isolated PR per item off `origin/master`, default-OFF + golden/cross-OS-gated, adversarial review before commit, full `run\_all\_tests.py`+ruff+mypy green, merge own green PRs, tracker+changelog in-PR, checkpoint memory. ? **shared checkout** (engine AND khelsutra ? sibling sessions commit concurrently; khelsutra needs a dedicated **worktree**: junction `web/node\_modules` then **`rmdir` the junction BEFORE `git worktree remove`**). ? **the LIVE alpha runs on the droplet** (SSH `~/ssh/khelsutra\_droplet` root) ? don't disrupt the `khelsutra-engine`/`cloudflared` services or the marketing site.

62

63 Prior session delivered the **engine compute/serving + doc-hygiene** track (10 PRs merged: #263?#267, #270, #272?#274, #276):

64 - **M2 FINALIZED on a real GPU** ? `worker train --trajectory-source detector` on a GCP **L4 (Singapore; Mumbai was stocked-out)** ? Gen-0 bundle `gs://khelsutra-rally-corpus/weights/0.1.0-l4/` (honest **n=2 plumbing** result, not a quality model). Both VMs **deleted**.

65 - **DigitalOcean DROPPED for compute ? GCP is the sole stack** (ADR-013; DO GPU not enabled/credit-funded, Paperspace subscription-gated). [[compute-stack-gcp-only]] · [[gcp-vm-always-delete]] · [[gcp-ops-agent-enable]].

66 - **DECOUPLED_COMPUTE delivered + ARCHIVED** ? `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

67 - **NEW tracked subproject: `docs/COMPUTE\_DECOUPLED\_SERVING/` + ADR-014** ? one pure core (DETECT?WINDOW?STITCH, never branches on profile) + `ComputeBackend`/`StorageBackend` profiles (**truly-local**, **cloud-serving** [Cloud Run GPU runs detect AND stitch, metered], **cpu-mock**). **? AWAITING OWNER M0 DESIGN SIGN-OFF before any code.**

68 - **Doc governance:** `docs/DOC\_STATUS.md` register + **archival due-diligence** (honestly-update?then-archive; codified in CLAUDE.md; [[doc-status-register]]); per-video docs reconciled; **data layer** carved into `docs/data\_pipeline/` (no-ML), field-ops/PMF?`archives/field-pmf/`, Roora?`archives/roora-vendor/`.

69

70 ? **Shared checkout** ? sibling sessions commit + flip branches here mid-session (hit twice this session). **Branch from `origin/master` explicitly, re-verify HEAD before commit, stage explicit paths, never `git add -A`**. [[gotcha-shared-checkout-branch-collisions]]. ? **`sed -i` over a glob churns CRLF on all matched files** ? separate real changes with `git diff --ignore-cr-at-eol` and revert the noise.

71

72 ## ? Next steps (resume here)

73 1. **? M0?M4 SHIPPED (#279/#280/#282/#283) ? the config/input/output/device SEAM LAYER is COMPLETE ? NOW M5+ (mostly owner-gated).** Decisions locked in README **\$2 D7?D15** (D7 scale-to-zero · D8 versioned DB pricebook USD/INR · D9 Cloudflare Access · D10 data-for-reels trade adults-only · D11 Gemini-ON-until-floor + D12/M11 flywheel · D13 user-switchable profiles · D14 NORTHSTAR gdrive-api OAuth = M12 · D15 suggest?confirm). **Engine build sequence:** ~M1 deployment.profile~ ? ~M2 input_backend~ ? ~M3 output-base~ ? ~M4 DeviceContext + CPU-fallback~ **ALL DONE.** **? M5** = truly-local profile e2e (preset + startup checks + first committed runlog) ? **? owner real-GPU** (needs a GPU box / owner's WSL machine to run detect+stitch locally, then commit the runlog). **M6/M7** Cloud Run dispatch + container image (? owner GCP spend), **M8** cost ledger + pricebook, **M9** edge-auth (Cf-Access), **M11** data-flywheel harness, **M12** gdrive-api OAuth. **Per-video-workspace P3 (`output/chunks\_\*)` is now subsumed by M3 ? reconcile/close it. **Claude-doable WITHOUT owner gating:** the M5 preset+startup-check code (the actual run is owner's); the M8 cost-ledger DB migration + pricebook + aggregation (pure code, owner does the real-cost calibration); the M11 harness plumbing. M6/M7/M9/M12 need owner cloud/counsel sign-off.

74 2. **Per-video workspace P1?P6** (#264 shipped P0): P1 v6 DB migration `workspace\_uri` ? P2 worker ? P3 `output/chunks\_\*` de-hardcode ? ? (Claude-doable, default-OFF).

75 3. **Ops-Agent codification into `deploy/gcp`** (auto-enable observability on every VM) ? small, [[gcp-ops-agent-enable]].

76 4. **[owner]** Grow the golden corpus (highest-leverage) · set ship-gate target #172 · full-corpus training (n?5) for real weights.

77 5. Full prioritized P0/P1/P2 list = `docs/NEXT\_STEPS.md` top.

78

79 ## ? Open decisions / owner gates

80 - **M0 design sign-off** (the blocker above) + the COMPUTE_DECOUPLED_SERVING \$8 questions.

81 - Doc-hygiene leftovers (`DOC\_STATUS.md` \$4): confirm the 3 COMPLETED top-level pointer-


```

stubs; `HOSTING_PLAN`/'`UI_PROPOSAL` currency.
82     - Housekeeping: retire the shared DO droplet `claude-do-rally-test` (~$24/mo) + the DO
token; rename repo off "badminton".
83
84     ## ? Ramp-up kit
85     - **Memory:** [[current-state]] (read first) . [[compute-stack-gcp-only]] . [[gcp-vm-
always-delete]] . [[gcp-ops-agent-enable]] . [[doc-status-register]] . [[gotcha-shared-checkout-
branch-collisions]].
86     - **Docs:**
`docs/COMPUTE_DECOUPLED_SERVING/{README,TESTING_STRATEGY,architecture.svg,runlogs/}` .
ADR-012/013/**014** in `docs/archives/decisions/DECISIONS.md` . `docs/NEXT_STEPS.md` .
`docs/DOC_STATUS.md` . `docs/PER_VIDEO_WORKSPACE.md` . `deploy/gcp/README.md` (GPU runbook;
teardown defaults to `delete`).
87     - **Code anchors (the pure core the serving project builds on):**
`backend/pipeline/detectors/base.py` (DetectorRunner CSV contract) .
`tracknet_runner.py::trajectory_to_action_windows` .
`backend/pipeline/stitcher/compiler.py::compile_highlights` .
`backend/pipeline/segmenters/trajectory_hybrid.py::_resolve_runner` (the compute seam) .
`backend/storage/*` . `backend/worker/__main__.py` . `backend/utils/workspace.py`.
88
89     ## ? Parallel track (NOT this session ? sibling sessions)
90     **KhelSutra brand-site + product UX/Hindi-Kannada localization** (repo
`Khelsutra/khelsutra` + #275 here) ? [[next-session-ux-localization]] / [[khelsutra-domain-
launch]]. The **rally-quality / owned-model** track is live too (R-series,
`RALLY_QUALITY_RESEARCH.md`; gated on corpus growth). These are background to the
compute/serving resume above.
91

```

6. assistant (2026-07-02T03:03:22.640Z)

The handoff confirms the droplet retirement at line 19. Now let me check the file's git history and whether any issue/commit already addresses #114 or the CF Access cleanup.

7. user (2026-07-02T03:03:25.455Z)

```

6e4dbaf docs(deploy): GPU_VM_SERVING.md ? move the live engine to an L4 GPU VM off the droplet
(#113)
7dc7de5 docs: freshness sweep ? reconcile 13 docs with the live repo (+ engine origin/master)
(#95)
7b04f02 deploy(kit): align with the live-verified droplet deploy (+ the gemini-default fix)
(#85)
3f0cdb8 docs(deploy): fix two would-break-the-deploy gaps in the alpha runbook
(RALLY_ALLOWED_HOSTS + gs:// jail) (#78)
4627830 docs(deploy): alpha deploy kit ? make Studio a shareable page (split: Pages + api.*
tunnel + CF Access) (#65)
---
114     OPEN     Cloudflare Access: an allowlisted email still couldn't sign in (alpha auth)
2026-06-23T10:25:52Z
112     OPEN     Share to WhatsApp sends only a text message, not the reel video bug
2026-06-22T18:27:52Z
111     OPEN     Resilient processing UX: survive page refresh + continually-updating progress
enhancement      2026-06-22T18:15:04Z
103     OPEN     SPA: show video stats (name, size, fps, resolution, duration) + persist them on
reload           2026-06-22T14:18:41Z
102     OPEN     SPA: &debug=1 (trace panel) / &debug=2 (full runlog) URL params for debugging +
tracing          2026-06-22T14:16:43Z
101     OPEN     SPA: show a friendly failure notice on a timeout/5xx (Cloudflare 524), not the
raw Cloudflare error HTML      2026-06-22T14:14:16Z
99     OPEN     SPA: keep the uploaded video's filename visible during processing (it disappears
mid-job)         2026-06-22T14:01:52Z
98     OPEN     SPA: replace the demo rally-selection the moment an upload/processing starts (so
the user knows their reel is coming)      2026-06-22T13:57:56Z
97     OPEN     Hosted alpha: split-origin (Pages app. + tunnel api.) breaks Cloudflare Access
for the SPA ? single-origin needed        2026-06-22T13:54:39Z
72     OPEN     i18n: add Telugu (te) locale (catalog + Noto Telugu font, key-parity green)

```

2026-06-21T18:17:34Z

8. user (2026-07-02T03:03:25.582Z)

No matches found

9. user (2026-07-02T03:03:56.043Z)

```
DEPLOY_ALPHA.md
GPU_VM_SERVING.md
cloudflared.config.example.yml
droplet
stage_from_bucket.sh
2026-06-29 14:46:49 +0530 docs(deploy): consolidate GPU_VM_SERVING.md ? rebase cleanup (#119)
```

10. user (2026-07-02T03:03:57.433Z)

```
14:> design hinge there is keeping the secrets (Gemini key / Access AUD / tunnel token) out of
plaintext
26:> stop it on the droplet" ? **no DNS change, no new Cloudflare token, no CF-Access change.**
37: - the **CF-Access AUD** + team domain ? already in the droplet's
`/srv/khelsutra/config.json`
87:/opt/khelsutra/venv/bin/pip install -q -e '[auth,storage-gcs]' # auth = the Cf-Access JWT
verify
123:and hosts are identical** to the droplet (it's the same tunnel + Access app).
203:curl -sI https://api.khelsutra.guru/api/health # 200 (or the 302?Access login if
unauthenticated ? that's correct)
212:DNS, the CF-Access app, and the dashboard ingress are unchanged ? only the connector host
moved.
218:- `https://app.khelsutra.guru` ? log in (email OTP) ? upload a match ? process ? pick
rallies ? **Build**
```

11. user (2026-07-02T03:04:13.926Z)

```
1      # Move KhelSutra Studio onto an L4 GPU VM (off the droplet)
2
3      Self-serve runbook to run the live engine (behind `api./app.khelsutra.guru`) on a **GCP
L4 GPU VM**
4      instead of the CPU droplet, and **cut the Cloudflare tunnel over** so traffic lands on
the VM. Lets
5      you switch detection between **Gemini** (cloud API, proven) and **WASB native** (uses
the L4 GPU).
6
7      This extends [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) (which deployed the same engine on the
droplet) ? it
8      reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the GPU-VM
scripts in the
9      **engine repo** (`badminton-highlight-indexer`: `deploy/gcp/`).
10
11     > **These are manual steps today.** Automating them with a one-shot **startup-script**
you feed into
12     > the GCP console (`create_gpu_vm.sh` already injects a metadata startup-script ? today
it only installs
13     > the Ops Agent) is backlogged: engine repo `docs/BACKLOG.md` ? *Batch / declarative
orchestration*. The
14     > design hinge there is keeping the secrets (Gemini key / Access AUD / tunnel token) out
of plaintext
15     > metadata.
16
17     > **Cost ??** A `g2-standard-4` (1xL4) bills **~$0.70?0.85/hr ? ~$500?600/month** while
it runs. You
18     > keep it up, so this is an ongoing cost. **`bash deploy/gcp/teardown.sh delete` is the
only way to
19     > stop billing** (a *stopped* VM still bills its disk). For bursty/intermittent load, a
```

```

scale-to-zero
20     > **Cloud Run GPU *service*** would be far cheaper (~$0 idle) ? that's the backlogged
"M7" and a bigger
21     > build; this VM is the fast path that works today.
22     >
23     > **Cutover is safe + reversible.** Bring the VM up and **verify it serving** before
touching the
24     > tunnel; the droplet stays as a hot fallback. The tunnel is **token-managed** (ingress
lives in the
25     > Cloudflare dashboard, not on the box), so moving it is just "run the same connector on
the VM, then
26     > stop it on the droplet" ? **no DNS change, no new Cloudflare token, no CF-Access
change.**
27
28     ---
29
30     ## 0. Prerequisites (gather these once)
31
32     - **gcloud** authenticated as the project owner (`gcloud auth login`) on the machine you
run this from.
33     - **GPU quota** `GPUS_ALL_REGIONS ? 1` and the SA Ops-Agent roles ? see the engine repo
34     `deploy/gcp/README.md` §2. (You already have L4 quota=1 from the nightly runs.)
35     - **The two secrets**, which you already have from the droplet deploy:
36     - the **rotated Gemini API key** (Drive: `gemini.api_key.txt`) ? only needed for the
Gemini path;
37     - the **CF-Access AUD** + team domain ? already in the droplet's
`/srv/khelsutra/config.json`
38     (`security.cf_access_aud` / `cf_team_domain`); reuse the same values.
39     - the **tunnel token** ? in the droplet's cloudflared unit (`systemctl cat
cloudflared` ? the
40     `--token` ? value). You'll copy it to the VM in §6.
41     - **For the WASB path only:** the weights at `gs://khelsutra-rally-
corpus/models/wasb_badminton_best.pth.tar`.
42
43     You'll work in **two repo checkouts** on your machine: this one (`khelsutra`) for the
config templates,
44     and the **engine repo** (`badminton-highlight-indexer`) for the GPU-VM scripts. Commands
below say which.
45

```

12. user (2026-07-02T03:05:33.187Z)

Structured output provided successfully